

JAVA PROGRAMMING – R25

Important Question Bank

5 important 2-mark questions + 2 important 16-mark questions for each syllabus topic/unit.

Based on the R25 / CS25C09 Java Programming syllabus covering Foundations of OOP and Java, Java Conceptual Foundations, Core Libraries, I/O & GUI, JDBC, and Collections.

UNIT I – Foundations of OOP and Java

Syllabus topics: Principles of OOP, JDK, Data Types, Variables and Arrays, Operators and Control Statements, Classes and Objects, Constructors, Method Overriding, Access Specifiers, Static Members, Inheritance

5 Important Two-Mark Questions

1. What is Object-Oriented Programming? List its major principles.
2. What are JDK, JRE and JVM? Explain their roles and relationship.
3. What is a class and an object? Explain class creation, object creation and methods.
4. What is method overriding? State its rules and illustrate it with a Java program.
5. What is inheritance? Name its types supported by Java.

2 Important Sixteen-Mark Questions

1. Differentiate between Java and C++. Explain the important features of Java with suitable examples.
2. Explain Java data types, variables, arrays, operators and control statements with suitable examples.

UNIT II – Java Conceptual Foundations and Capabilities

Syllabus topics: Packages and Interfaces, Exception Handling, Multithreaded Programming, Enumerations, Autoboxing and Annotations, Generics, Lambda Expressions

5 Important Two-Mark Questions

1. What is a package? Why are packages used in Java?
2. What is an interface? How is it different from a class?
3. What is an exception? Differentiate checked and unchecked exceptions.
4. What is a thread? List the states of a thread.
5. What are generics? What is autoboxing?

2 Important Sixteen-Mark Questions

1. Explain packages and interfaces in Java. Show how user-defined packages are created and imported.
2. Explain interfaces and how Java achieves multiple inheritance through interfaces, with a suitable program.

UNIT III – Java Core Libraries

Syllabus topics: String Handling, java.lang package, Utility Classes, java.io package, Event Handling, AWT Controls, Layout Managers

5 Important Two-Mark Questions

1. What is a String in Java?
2. What is the purpose of the java.lang package? Name important classes.
3. What is file I/O? Differentiate byte streams and character streams.
4. What is an event? What is an event listener?
5. What is a layout manager? Name its types.

2 Important Sixteen-Mark Questions

1. Explain String, StringBuffer and StringBuilder. Discuss immutability and commonly used string methods.
2. Explain important java.lang classes and utility classes with suitable examples.

UNIT IV – I/O and GUI Programming in Java

Syllabus topics: Console I/O, Files and File Handling, Swing, Menus, Swing Event Handling

5 Important Two-Mark Questions

1. What is console I/O in Java?
2. What is a file? Name commonly used file-handling classes.
3. What is Swing? Mention its advantages.
4. What is a menu in Swing? Name its main components.
5. Differentiate between AWT and Swing.

2 Important Sixteen-Mark Questions

1. Explain different methods of reading and writing console input/output in Java with examples.
2. Explain file creation, reading, writing and appending in Java with a suitable program.

UNIT V – Database Connectivity Using Java

Syllabus topics: JDBC, Types of JDBC Drivers, JDBC Packages, JDBC API and Driver, JDBC/ODBC Bridge, Database Connections, Database Operations

5 Important Two-Mark Questions

1. What is JDBC? Expand JDBC.
2. List the types of JDBC drivers.
3. What are JDBC packages? Name important interfaces/classes.
4. What is the JDBC/ODBC bridge?
5. What is a database connection?

2 Important Sixteen-Mark Questions

1. Explain JDBC architecture and the steps involved in establishing a database connection and executing SQL operations.
2. Explain the different types of JDBC drivers and compare their characteristics.

UNIT VI – Java Collection Framework

Syllabus topics: Collection Framework, Regular Expressions (Regex), java.util package, Arrays and Collections, Map Interfaces, Iterators, Generics

5 Important Two-Mark Questions

1. What is the Java Collection Framework?
2. What is an ArrayList?
3. What is a Map interface? Give examples.
4. What is an Iterator?
5. What is a regular expression?

2 Important Sixteen-Mark Questions

1. Explain the Java Collection Framework hierarchy and compare List, Set, Queue and Map interfaces.
2. Explain ArrayList operations with a suitable Java program and compare ArrayList with arrays.